

JavaScript Array Datatype - Theory Notes

1. What is Array?

Array stores multiple values in a single variable.
Values are stored in ordered positions called indexes.

2. Syntax

```
let arr = [10, 20, 30];
```

3. Important Note

Array index starts from 0.
First value = arr[0]

4. Typeof

typeof array -> object
Array is a special kind of object.

5. Common Operations

push() add end
pop() remove end
shift() remove first
unshift() add first

6. Useful Methods

length, indexOf(), includes(), join(), slice(), splice(), reverse(), sort()

7. Mixed Values

Array can store numbers, strings, booleans, null, objects.

8. Nested Array

An array can contain another array.

9. Looping

Use for loop or for...of loop.

10. Real Uses

Student names, product lists, marks, tasks, cart items.

11. Best Practice

Use `Array.isArray(value)` to check array.

12. Quick Revision Table

Feature	Example
Create Array	<code>[1,2,3]</code>
Access First Item	<code>arr[0]</code>
Length	<code>arr.length</code>
Check Array	<code>Array.isArray(arr)</code>
Typeof	<code>object</code>